MTH2008 Real Analysis: Graded Formative Assessment 2

This Graded Formative Assessment Sheet covers the material presented in Weeks 3, 4, 5 and 7.

If you would like to receive feedback on your work, please hand in solutions to Questions 5 and 10 by 12:00 noon on Wednesday 11 November 2026. As it is optional to submit solutions to these two assigned questions for feedback, if your submission is late, it will be at your tutors discretion as to if it is graded or not.

Note yours solutions to the two assigned questions, in total, should not exceed 750 words, and that this is not a target. Indeed, if you find yourself writing more than this, take a step back and think about your solution and your approach to the problem.

Each of the two assigned questions will be graded on a 10-point scale and your mark on ELE will be scaled by a factor of 5 so as to be a mark out or 100.

Please also ensure you upload your solutions to ELE in a single PDF (Portable Document Format) unless otherwise instructed by your tutor. As it is optional to submit solutions to the two assigned questions for feedback, if your file is not uploaded in a single PDF, it will be at your tutors discretion as to if it is graded or not.

Note also, that Summative Assessment 2 will only consist of questions taken directly from this assessment.

Learning outcomes for Graded Formative Assessment 2

Functions of a single variable, continuity and limits, uniform continuity and differentiable functions of a single variable.

Assessment criteria for Graded Formative Assessment 2

If this were an assessment which counted towards your final grade, to achieve a pass mark, you will have needed to demonstrate that you know the basic definitions of the subject and can verify whether or not they hold in some familiar cases. To achieve full marks, you will have needed to demonstrate a thorough understanding of the concepts by relating them to more elaborate mathematical contexts. You will also have needed to demonstrate the ability to construct mathematical arguments and to communicate them effectively through clearly explained and logically correct proofs with accurate use of appropriate mathematical notation and terminology.

Statement on use of GenAI tools for Graded Formative Assessment 2

As Graded Formative Assessment 2 has the option for you to submit your solutions to a selection of questions (see above for precise details) in order for you to gain feedback, it is advised that you treat this assessment with the classification of being AI-Minimal. This means that you may use AI tools for checking spelling and grammar mistakes only, with no other impact on the structure or content of the assessment. This is because using AI tools outside of these uses prevents fair assessment of your ability to achieve the module learning outcomes. As such, when writing your assessment, try not to use AI tools in the following ways:

(i) for uses other than checking your spelling and grammar,

(ii) to translate more than a word or short phrase into English,

(iii) to upload sensitive or identifying material to an AI tool, and

(iv) to present material that has been generated by AI as your own work or the work of someone else.

1. Let $f : \mathbb{R} \to \mathbb{R}$ be defined, for $x \in \mathbb{R}$, by $f(x) = \mathrm{e}^x$. Determine when $\lim_{x\to\infty} f(x)$ exists, justifying your answer using the definition of limits at infinity.

2. For α a positive real number, let $f_\alpha : \mathbb{R} - \{0\} \to \mathbb{R}$ be defined, for $x \in \mathbb{R} - \{0\}$, by $f_\alpha(x) = |x|^{-\alpha} \sin(x)$. Determine when $\lim_{x\to\infty} f(x)$ exists, justifying your answer using the definition of limits at infinity.

3. Using either the definition of continuity, or known results on addition, products and compositions of continuous functions, determine, justifying your answer, all $x \in \mathbb{R}$ at which the function $x \mapsto \sqrt{1 - x^2}$ is well-defined and continuous.

4. Consider a function $f : \mathbb{R} \to \mathbb{R}$ with the following property: If $g : \mathbb{R} \to \mathbb{R}$ is any function for which $\lim_{x\to x_0} g(x)$ does not exist for some $x_0 \in \mathbb{R}$, then $\lim_{x\to x_0}(f(x) + g(x))$ also does not exist. Prove that $\lim_{x\to x_0} f(x)$ must exist. You may use, without proof, standard results established in lectures, including the algebra of limits and the evaluation of standard limits.

5. Let I be an interval in $\mathbb{R}$. Let $f\colon I \to \mathbb{R}$ be a continuous function such that $f(x)^2 = 1$ for all $x \in I$. Prove that $f(x)$ is constant and either $f(x) = 1$ for all $x \in I$ or $f(x) = -1$ for all $x \in I$.

6. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that there exists $\lambda \in (0, \infty)$ with $f(x + \lambda) = f(x)$ for all $x \in \mathbb{R}$. Prove that f attains its maximum and minumum.

7. State an example of a function that is continuous and bounded on $\mathbb{R}$, but not uniformly continuous.

8. Give a justification for your answer to Question 7.

9. Suppose that f and g are uniformly continuous functions on an interval $I \subseteq \mathbb{R}$. Determine if $f + g$ is uniformly continuous on I. Justify your answer.

10. Suppose that f and g are uniformly continuous functions on $\mathbb{R}$. Determine if $f \cdot g$ is uniformly continuous on $\mathbb{R}$. Justify your answer. Note, you may use known functions which are uniformly continuous and which are not uniformly continuous without giving a proof of uniform/non-uniform continuity.

11. Let $f : \mathbb{R} \to \mathbb{R}$ be defined, for $x \in \mathbb{R}$, by

$$f(x) = \begin{cases} x^2 \cdot \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

(i) Prove that f is differentiable on $\mathbb{R}$.

(ii) Prove that it's derivative is not continuous at 0.

You may use, without proof, standard results established in lectures, includes the differentiability of standard functions such as trigonometric functions and polynomials, the algebra of limits, the standard rules of differentiation, and the evaluation of standard limits.

12. Suppose that $f : \mathbb{R} \to \mathbb{R}$ is such that $f'(0)$ exists and $f(x + y) = f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$. Prove that the derivative of f exists at all $x \in \mathbb{R}$.